

Teleoperation of Semi-Autonomous Vehicles: Technologies and Human Ergonomic Perspectives

Ahmed Alshehhi, Khalid Al Johani, Burak Kizilkaya *Member IEEE*, Olaoluwa Popoola *Senior Member IEEE*,
Dezong Zhao, Muhammad Ali Imran *Fellow IEEE* and Shuja Ansari *Senior Member IEEE*

Abstract—In an era of increasing automation, teleoperation has emerged as a critical enabler for extending autonomy into complex or high-risk environments where full automation remains infeasible. Although the long-term goal of fully autonomous road vehicles continues to drive research, current technological limitations require robust systems that ensure reliability, safety, and adaptability. Teleoperation addresses this need by allowing human operators to remotely monitor, guide and directly control vehicle actions under conditions where autonomous decision-making falls short. Motivated by the need to bridge the gap between autonomy and human oversight, this survey provides a comprehensive review of the field focusing on four foundational pillars: key communication technologies and network performance metrics, control methodologies in teleoperation, autonomy capabilities under teleoperation constraints and human ergonomics. Furthermore, we synthesize recent research trends and identify open challenges.

Index Terms—Teleoperation, Communications, Control, Semi-Autonomous Vehicles, Latency Mitigation, Artificial Intelligence

I. INTRODUCTION

TELEOPERATION of semi-autonomous vehicles has emerged as a pivotal technology in industrial and vehicular applications [1], enabling human operators to remotely control or supervise ground vehicles while benefiting from automation. This approach is particularly valuable in hazardous environments where full autonomy remains impractical due to technical or safety constraints. Teleoperation extends the capabilities of autonomous ground vehicles (AGV) by integrating human decision-making with machine efficiency. The effectiveness of such systems, as shown in Figure 1, is determined by the interplay between four core components: communications, control, autonomy and human factors.

Effective communication forms the backbone of teleoperation. The transmission of control commands and sensory feedback must occur with minimal latency and sufficient bandwidth to support responsive and intuitive control [2]. Traditional wireless networks often fall short in meeting these real-time requirements, especially when subjected to interference, congestion or geographic limitations. However, recent advances in wireless communications such as the rollout of 5G [3] and the development of ultra-reliable low-latency communication (URLLC) [4] protocols promise to greatly enhance the responsiveness and reliability of

teleoperated systems.

The control systems underpinning teleoperation must be both robust and adaptive [5]. Operators often rely on feedback streams such as video, LiDAR or proprioceptive sensors to make informed decisions [6]. Control algorithms must translate these decisions into real-time actuation commands, compensating for communication delays and ensuring stability under uncertain dynamics. Model predictive control (MPC) techniques [7] combined with machine learning enhancements such as gated recurrent units (GRUs) [8] have recently shown promise in addressing the time-delay challenges inherent in teleoperation. These hybrid approaches enable predictive compensation for latency while preserving control stability and human interpretability.

Human factors in semi-autonomous teleoperation systems include pivotal roles such as trust, cognitive workload and situational awareness where they are critically affected by network performance and interface design [9]–[11]. We explore how communication impairments such as latency and bandwidth limitations influence operator behavior and system safety in both single and multi vehicle teleoperation contexts [12]–[14]. The paper also compares immersive VR based interfaces with traditional screen based modalities [15]–[17], highlighting trade offs in presence, workload, and decision making. Finally, we review emerging training strategies including immersive simulations and adaptive feedback systems that aim to improve operator readiness, trust calibration, and handover performance under real time constraints [18]–[20]. This integrated perspective offers a human centered framework for understanding and improving the teleoperation of semi autonomous vehicles across diverse applications.

Numerous surveys have explored isolated domains of vehicle teleoperation, a comprehensive review that spans communication, control, autonomy and human ergonomic perspectives in semi-autonomous teleoperation remains underexplored. Teleoperation architectures have been categorized into direct, supervisory and shared control modes, and their data requirements and operator workload compared but this work focuses primarily on control frameworks [21]. Sources of network latency in CAV teleoperation have been systematically assessed with edge-computing and packet-scheduling strategies proposed to mitigate delays, yet these findings are not integrated with control stability or interface design considerations [2]. A detailed taxonomy

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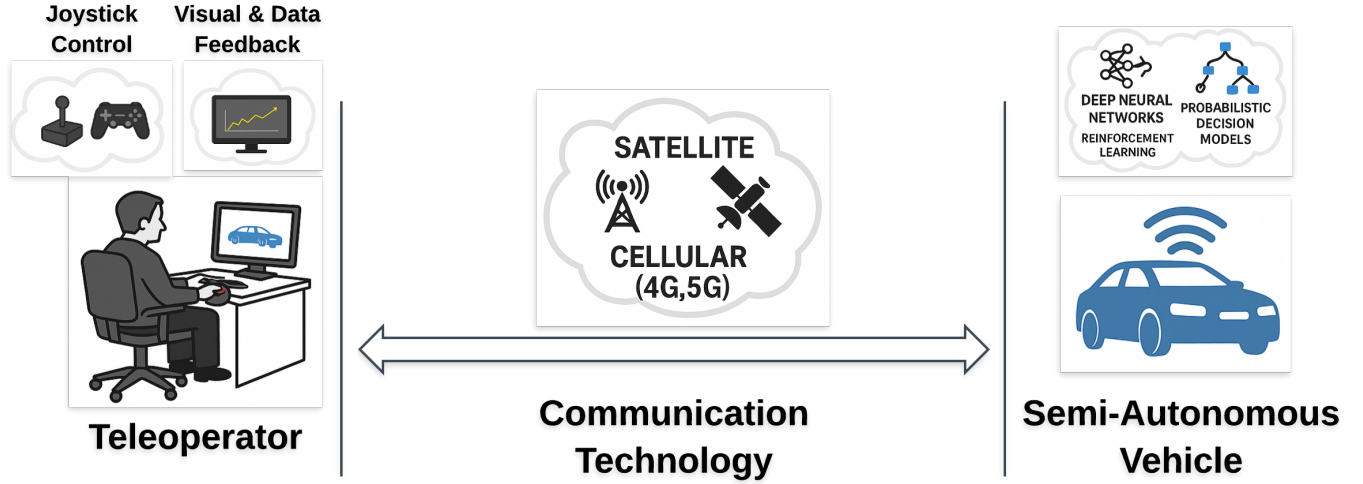


Fig. 1. Illustrates how teleoperators interact with semi-autonomous vehicles through various communication technologies. Operators issue commands and receive both visual and haptic feedback through a dedicated user interface.

of haptic bilateral teleoperation systems analyzes device ergonomics, bandwidth requirements and force-feedback performance, though it excludes video-centric interfaces and higher-level autonomy layers [22]. In addition, a broad overview of teleoperation enhancement techniques covering video compression schemes, adaptive streaming protocols and predictive control via AI for general mobile robots remains focused on low-level robotics rather than vehicle-specific teleoperation challenges [23]. While each of these works contributes valuable, specialized insights, there remains no unified survey integrating the four interdependent dimensions essential for robust semi-autonomous vehicle teleoperation. This paper reviews the existing research within four domains: Communication, Control, Autonomy and Human Factors, analyzing their inter-dependencies and challenges within semi-autonomous teleoperation frameworks. Figure 2 shows the structure of the sections and subsections of this survey.

In this survey paper, our primary contributions are as follows:

- We systematically evaluate cellular, Wi-Fi and V2X solutions for teleoperation, offering a critical comparison of their latency, bandwidth and coverage trade-offs. This analysis provides a structured framework to guide communications design for reliable teleoperated vehicle systems.
- We present an in-depth survey of control methodologies, including time-delay compensation, shared/assistive control and hybrid AI-theoretic approaches. Our contribution lies in synthesising classical and learning-based techniques within a unified perspective tailored to teleoperation challenges.
- We define and analyze layered autonomy architectures that enable predictive interventions and supervisory control. Our work offers design guidance on balancing human authority with autonomous reliability in semi-

autonomous vehicle systems.

- We analyze operator interface modalities including video, haptics and virtual reality alongside cognitive workload and multi-feed management issues. Our perspective contributes operator-centred design principles aimed at improving situational awareness, reducing cognitive burden and enhancing system safety.

II. COMMUNICATION

Reliable communication is fundamental to teleoperation, as the quality of data transmission directly affects control precision and response times [5]. In a typical teleoperation system, a human operator at a control station sends commands to a distant vehicle and receives feedback like video, audio or haptic signals from the vehicle over a communications network [24]. This capability allows vehicles to benefit from human judgment in edge cases while otherwise operating autonomously. To enable effective teleoperation, robust communication is paramount [25]. The communication taxonomy can be classified based on network type, latency sensitivity, and resilience mechanisms.

A. Communication Technologies

Several communication technologies facilitate teleoperation, each with varying degrees of latency, reliability and bandwidth. Table I summarizes the latency and coverage of common communication technologies.

1) Dedicated Short-Range Communications (DSRC):

DSRC refers to wireless communication standards based on IEEE 802.11p known as Wireless Access in Vehicular Environments (WAVE) [26]. DSRC was originally developed for vehicle-to-vehicle and vehicle-to-infrastructure safety messages, offering low latency (1–10 ms) in a short range typically from a few hundred meters and up to 1 km in

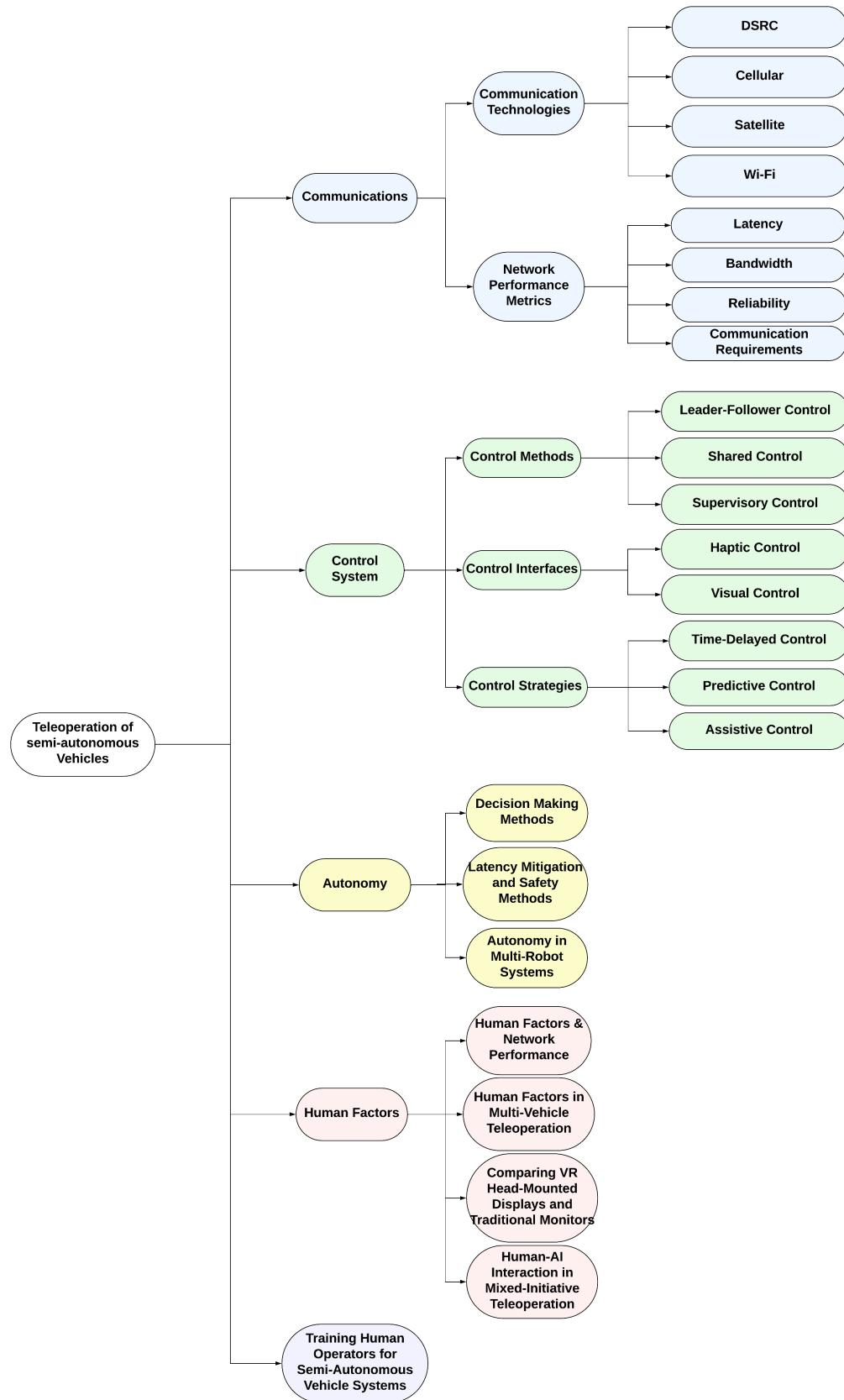


Fig. 2. Shows a summary diagram of the structure of the survey

ideal conditions. In teleoperation [27], DSRC could be used for direct vehicle-to-operator (V2O) links if the operator is nearby or for connecting to roadside units that forward data. It provides fast, localized communication. For example, an operator in a follow vehicle could directly control a lead vehicle via DSRC. However, DSRC's limited range and requirement for line-of-sight make it unsuitable for long-distance or non-line-of-sight teleoperation [28]. Standards bodies like the European Telecommunications Standards Institute (ETSI) have defined DSRC protocols for vehicular use [29], but the technology faces competition from cellular V2X. In summary, DSRC offers ultra-low latency and does not depend on carrier networks. DSRC operates in a self-contained band reserved for intelligent transport systems, using direct ad hoc links between vehicles and roadside units rather than routing data through a mobile operator's cellular infrastructure. However, its geographic coverage is very restricted.

2) *Cellular Communications:* Cellular networks are currently the backbone for most long-range teleoperation systems [30]. 4G LTE networks provide widespread coverage and broadband data rates where it has been noted that 4G LTE is the best candidate for vehicle teleoperation today due to its ubiquity. LTE can typically offer tens of Mbps throughput and latencies around 50–100 ms under good conditions [31]. However, 4G may struggle with consistency, a study reported [27] median latencies around 100 ms for LTE in vehicular scenarios, and latency can spike if the network is congested. 5G networks substantially improve capabilities in teleoperation. 5G ultra-reliable low-latency communications (URLLC) targets one-way latencies on the order of 5 ms or less, suitable for real-time control [32]. Data rates in 5G are also dramatically higher (peak up to 1–10 Gbps) compared to 4G (which offers up to 100 Mbps–1 Gbps). These enhancements enable higher fidelity video feeds and more responsive teleoperation [33]. Additionally, 6G is envisioned to exploit terahertz spectrum and AI-native network orchestration to deliver data rates on the order of hundreds of gigabits per second with end-to-end latencies potentially below one millisecond. This leap will allow teleoperation systems to stream ultra-high-definition video, real-time haptic feedback and rich sensor data virtually without perceptible delay [34]. Advanced network slicing and dynamic resource allocation will ensure ultra-reliability and consistent low-latency performance even in highly congested or mission-critical scenarios [35].

3) *Satellite Communications:* In areas lacking network coverage like rural roads or ocean shipping routes, satellites can link a teleoperated vehicle to its operator [36]. Traditional geostationary (GEO) satellites, however, introduce very high latency on the order of 500–600 ms one-way (approximately 1200 ms round-trip) due to the 36,000 km orbital distance [37]. This latency is generally considered too large for safe real-time driving far exceeding 200–300 ms maximum tolerable round-trip delay for smooth teleoperation [38], [39]. GEO satellites do offer broad coverage to entire continents

and decent downlink bandwidth where satellite systems can provide downlink capacities up to 1 Gbps, though uplinks may be limited to a few Mbps. Newer Low-Earth Orbit (LEO) satellite constellations like Starlink orbit at 500–1200 km and can reduce latency dramatically, into the tens of milliseconds [40]. For example, a LEO satellite at 600 km has a propagation delay of 2–4 ms to the ground, and even accounting for network routing, Starlink reports 20–40 ms latencies comparable to terrestrial cellular [41]. LEO satellites thus hold promise for teleoperation by providing near-global coverage and low latency. The downside is that many satellites are needed for continuous coverage and current user terminals require clear sky view. In summary, satellite communication can extend teleoperation to remote areas and ensure connectivity redundancy where satellite can be used as a backup to ground networks, at the cost of higher latency and substantial deployment expense. Hybrid approaches are being studied to seamlessly integrate LEO satellite links with 5G and 6G networks [42].

4) *Wi-Fi:* Wi-Fi is widely used for local wireless connectivity and can play a role in teleoperation, particularly in confined environments like warehouses, ports, or mines [43] [44]. Wi-Fi can provide very high data rates up-to hundreds of Mbps up to Gbps for modern standards but over limited range typically 50–200 m [45]. In teleoperation testbeds and research, Wi-Fi links (or wired LAN connections) are common for connecting a vehicle to a nearby operator station [2]. For example, an autonomous forklift in a warehouse might be teleoperated via a private Wi-Fi network covering the facility. Latency on Wi-Fi is usually low in the order of tens of milliseconds or less on an uncongested network but can spike if the spectrum is crowded or if there is interference [46]. An advantage of Wi-Fi is that it operates in unlicensed bands and can be set up privately with full control, without relying on telecom providers. However, Wi-Fi does not natively support long distances or high mobility. Therefore, Wi-Fi is well-suited for on-site teleoperation where controlled settings like teleoperating construction machinery on a worksite or mining trucks in an open pit mine where a local wireless LAN can be deployed [47].

B. Network Performance Metrics

Ensuring effective teleoperation requires meeting stringent network performance criteria. The following metrics are identified in the literature as particularly critical:

1) *Latency:* Network latency is perhaps the most crucial metric for teleoperation. It is well-established that human operators can only tolerate limited delay between their action (e.g. a steering input) and the observable response (e.g. video update showing the vehicle turning) [52]. Table I shows the typical latencies for common communication technologies. If latency is too high, the vehicle becomes difficult or impossible to control in real time [2]. Studies have shown that latency as low as 10–20 milliseconds one-way can start to noticeably affect an operator's performance [53] [54]. Under about 100

TABLE I
ILLUSTRATES THE CURRENT COMMUNICATION MODES USED IN TELEOPERATION.

Communication Technology	Typical Latency (ms)	Coverage
Fiber Optic (Wired) [48]	1-5	-
5G [49]	1-10	Several Km
4G LTE [50]	30-50	5-50 Km
Wi-fi [51]	10-30	100m
V2X Communication [48]	1-10	-
Satellite [48]	500-800	Thousands of Km

ms, effects are minor and can be mitigated with training, but beyond that the driver must consciously adapt. Around 200–300 ms total (round-trip) latency is often cited as the upper bound for smooth teleoperation of a car at city speeds [21]. Experiments show that at 500 ms delay, driving is still possible if the delay is consistent where the operator can compensate for a fixed lag. However, as latency approaches 1s, real-time control breaks down where one study noted teleoperation became largely infeasible beyond 1000 ms (1s) [55]. In summary, ultra-low latency communication is essential and the goal is usually to keep one-way latency below 50 ms (ideally <10 ms) for teleoperated driving. Achieving low latency involves not just fast wireless links hence interest in 5G/6G, but also optimized network routing and fast encoding of sensor data [56]. Even perception can be affected by small delays. For example, haptic feedback delays over 20 ms can reduce an operator’s ability to feel and react to forces [57]. Thus, minimizing latency across all modalities is a top priority for teleoperation systems.

2) *Bandwidth*: Teleoperation is a data-intensive application where a remote driver typically requires multiple video feeds (forward camera, side/rear cameras or 360° view), audio, and other telemetry [6]. A single 720p video stream can consume 1–5 Mbps, and higher resolutions or multiple streams scale this up. For instance, streaming uncompressed high-definition video can easily exceed 100 Mbps, which is impractical over networks; therefore video is usually compressed at the cost of some latency [58]. A study in teleoperated driving [59] highlights that available bandwidth can be a limiting factor if the network cannot sustain the needed data rate, video quality will drop or frames will be skipped, impairing the operator’s view. One approach achieved acceptable remote driving at only 1.25 Mbps by splitting the video into critical and non-critical portions, showing that smart compression can reduce bandwidth needs [60]. Nonetheless, ample bandwidth is highly desirable where high bitrates allow low-latency, high-quality video and multi-sensor data. Network technologies differ greatly in bandwidth (e.g. 5G > 4G > DSRC). A rule of thumb is to budget several Mbps per camera feed for reliable teleoperation. Uplink bandwidth (vehicle to operator) is especially important to send video and LiDAR out from the vehicle. For example, one teleoperated driving trial over LTE [61] had to increase the uplink rate limit to 1.5 Gbps using LTE-Advanced features to carry multiple video streams and control data with low overhead. In summary, insufficient bandwidth can choke a teleoperation system causing pixelated video or delayed control. Therefore, network capacity is a

key performance metric. Modern solutions, like 5G and 6G with network slicing aim to guarantee a certain bandwidth for teleoperation even in congested networks.

3) *Reliability*: In teleoperation, losing packets especially control commands or key video frames can be dangerous [62], [63]. A dropped packet might mean a missed brake command or a momentary freeze in the video feed. Teleoperation research [64] emphasizes that communication disruptions or high packet loss can detach the operator from the vehicle. For safe operation, the network should deliver packets with minimal loss and ensure near 100% uptime. Reliability is typically quantified by the packet loss rate which should be as low as possible (ideally <0.1%) and by the connection availability which is defined as the percentage of time the link is active [65], [66]. For teleoperation, redundancy is often used to boost reliability. For example, sending duplicate critical packets or using multiple network links. A study notes [67] that packet loss significantly impacts teleoperation but are less studied. For instance, a network might have low average latency but if it frequently drops packets, the effect on a remote driver’s ability to control smoothly can be severe where the vehicle appears to not respond to inputs during dropouts. Thus, reliable transport protocols or application layer error correction are important. In summary, a teleoperation network must maintain a stable, error-free connection, as even brief interruptions can lead to hazardous situations.

4) *Communication Requirements*: In a communication network, communication must be secure without compromising real-time performance. Teleoperated vehicles present a significant cybersecurity risk [68], [69]. If an attacker intercepts or injects commands, they could effectively hijack the vehicle. Man-in-the-Middle (MITM) and false-data-injection attacks enable adversaries to intercept or alter critical steering or braking commands in teleoperated vehicles, effectively hijacking motion control and inducing unsafe maneuvers [70], [71]. Denial-of-Service (DoS) and jamming attacks can sever low-latency communication links, causing control freezes or latency spikes that destabilize closed-loop control and elevate collision risk [72], [73]. Thus, strong encryption and authentication of teleoperation data is mandatory. Researchers have begun examining the trade-off between security and latency [74]. For instance, how much encryption can be added before control performance is affected. One study on space teleoperation [75] noted that enabling command authentication and encryption added a small delay, but this was considered

acceptable given the high stakes of malicious interference. Modern networks like 5G have built-in encryption at the radio layer [76], yet end-to-end encryption at the application layer may be needed for full security. Standards and guidelines like ISO/SAE teleoperation security standards [77], [78] are being developed to ensure teleoperation communication is “unhackable”. For this review, the key point is that security must be achieved in tandem with low latency. Techniques like lightweight cryptography or even physical-layer security are explored [79] to secure teleoperation without slowing it down. From an operator’s perspective, a secure link should be indistinguishable from an insecure one in performance. Security is thus an essential non-negotiable requirement in teleoperation network design.

III. CONTROL

A. Control Architecture

1) *Leader-Follower Control*: In Leader-Follower Control or direct teleoperation, the human operator has complete control over the vehicle’s actions in real time [80]. Commands from a human-machine interface such as a steering wheel or joystick are mapped directly to the vehicle actuators with no intermediate autonomy. This teleoperation mode benefits from human decision-making and intuition. The operator must rely on remote feedback (typically video) to react, which can be challenging if sensor views are limited or delayed [81]. Studies note that direct teleoperation lacks any intelligent assistance to the user [82]. As a result, performance can degrade under high latency or poor visibility, and operators may experience fatigue or reduced situational awareness. Nonetheless, direct control is crucial for precise maneuvers when full human judgment is required, e.g., teleoperating in complex or dynamic environments. Research in this method often focuses on improving the operator’s telepresence and reducing workload through better interfaces [83]. Additionally, safety measures such as remote emergency braking or automatic collision avoidance layers are also introduced in practice to complement direct tele-driving [82]. A key challenge moving forward is how to maintain high levels of human control while preventing operator errors.

2) *Shared Control*: Shared control is a hybrid approach where the human and an autonomous system simultaneously contribute to vehicle control [84]. The vehicle assists the user in accomplishing the task rather than simply executing raw user commands. In practice, the operator might provide high-level inputs (e.g., desired direction or speed) while the autonomy fine-tunes or overrides actions to ensure safety and task efficiency. For example, the system may integrate the user’s steering commands with sensor information about obstacles, modifying the vehicle’s trajectory to avoid collisions [85]. The advantage of shared control is improved safety and ease of operation where the vehicle can prevent many human errors (e.g., by refusing to drive into an obstacle or by smoothing jerky commands) and reduce operator workload during challenging maneuvers [86]. In fact, comparative experiments have found that shared

control schemes can increase task performance (e.g., fewer collisions or faster completion) compared to purely manual teleoperation [87]. The main limitation of shared control lies in the potential ambiguity and human-machine conflict where the operator must understand and trust the vehicle’s assistance [88]. Loss of direct control authority can cause confusion or reduced situational awareness if the system intervenes without effectively communicating its actions. Shared control teleoperation is being explored in real-world applications such as driver-assistance for remote parking or lane-keeping, where the remote operator is still in charge but advanced driver-assistance systems (ADAS) on the vehicle help ensure safety [89].

3) *Supervisory Control*: In supervisory teleoperation, the human operator does not continuously drive the vehicle but instead monitors the autonomous vehicle’s operation and intervenes only at a high level when needed [21]. Here, most control functions are handled by the vehicle’s onboard automation, and the operator primarily issues strategic or goal-level commands or takes over in emergencies [90]. On-board automation is realized through layered sensor-fusion and perception modules. These systems combine camera, LiDAR and radar measurements using probabilistic filters to generate real-time environmental models. The resulting models feed motion-planning algorithms, which compute safe trajectories autonomously [91]. In an ideal supervisory control mode, the operator might specify a destination, approve an autonomously generated path or simply oversee multiple vehicles, while the vehicle’s local control loops execute the low-level driving tasks autonomously [92]. The benefit of supervisory control is efficiency: one operator can supervise several semi-autonomous vehicles (e.g., a fleet of driverless taxis or delivery robots) and only needs to step in when the automation encounters a situation it cannot handle. Researchers argue that a centralized human supervisory center is necessary as an exception-handling layer to ensure safety in uncertain scenarios that an automated driving system (ADS) cannot easily solve on its own [93]. The challenge in supervisory teleoperation is maintaining the human’s situation awareness and readiness to intervene despite long periods of passive monitoring. It is well documented that continuous monitoring with little action can lead to operator underload and vigilance drop, slowing reaction times when an intervention is finally required [94].

B. Sensing Modes

1) *Haptics*: Haptics involve the use of force-feedback or tactile cues on the operator’s control devices to convey information or enhance control [22]. In tele-driving, this often means a force-feedback steering wheel or joystick that can resist the operator’s movements or vibrate to signal cues. Research has shown that haptic feedback is highly useful for collision avoidance and guidance [95], [96]. For example, a bilateral shared-control system may apply a repulsive force on the steering wheel when an obstacle is near, nudging the operator away from a collision path. This essentially allows the vehicle to communicate dangers through touch, not just

through visual warnings. Haptic feedback can also guide the operator along a desired trajectory; one study provided force and vibration cues to help an operator follow a predefined path with a teleoperated device [85]. The advantages of haptic control include improved operator awareness of the remote vehicle's state and environment, as well as reduced cognitive load. Some decisions are implicitly shared by the system applying forces, rather than the operator needing to notice every hazard visually [97]. Implementations like a predictive haptic steering feedback have been shown to help maintain safe lateral control in teleoperated driving, warning the user of upcoming turns or lane departures via forces on the wheel [57]. However, there are limitations. Haptic devices must be finely tuned where too much force or incorrect cues can confuse the operator or even destabilize the control. There's also a technological overhead. Equipping the control station with high-quality force feedback systems like steering actuators or vibrotactile seats adds cost and complexity [96].

2) *Visual*: Visual input is the cornerstone of teleoperation control, nearly all teleoperated vehicle systems rely on video feeds from the remote vehicle's cameras to the human operator [98]. The operator's situational awareness depends heavily on the quality and timeliness of these video streams [99]. High-resolution, wide field-of-view video with minimal latency allows the remote driver to approximate an in-vehicle experience. Conversely, poor visual feedback can severely impair performance. The study [100] quantitatively showed that image quality has a significant influence on teleoperation task performance and reaction times. Lower resolution or frame rate (due to bandwidth limits) leads to slower hazard recognition and responses by the operator, directly impacting safety. Likewise, narrow camera field-of-view and lack of depth perception (2D displays) increase cognitive workload and can even cause motion sickness or disorientation in operators [101]. To mitigate such limitations, researchers have explored improved visual interfaces: multiple camera angles, 360-degree video, stereo or 3D displays, and even virtual reality (VR) or augmented reality (AR) headsets to immerse the operator [102], [103]. The goal is to approach the richness of real in-person driving vision. The advantage of visual control is its intuitiveness. Humans naturally rely on sight for driving, so providing a live video feed leverages the operator's existing skills. Virtually all real-world teleoperation deployments (e.g., remote-controlled cars, delivery robots) use video as the primary feedback modality because it conveys rich environmental detail needed for driving. The limitations, as noted, center on bandwidth, latency, and human factors. A teleoperator must divide attention between possibly several video monitors and instrument readouts, which can be demanding [48].

C. Control Strategies

1) *Time-Delayed Control*: Dealing with time delay is arguably the central technical challenge in teleoperation [104]. Unlike in direct on-board control, a teleoperator's commands and feedback signals travel over networks

that introduce latency. Depending on the distance and communication technology, this delay can range from a few milliseconds for local networks to hundreds of milliseconds in cellular networks or even seconds in satellite links. Time-delayed control refers to strategies that allow stable and effective teleoperation despite these delays [105]. The impact of latency on teleoperation performance is well documented where humans can detect delays as low as 10–20 ms [106], and a delay around 300 ms is enough to noticeably impair driving performance causing overshoot. When delays reach 1 second or more, real-time tele-driving becomes extremely difficult where operators must adopt open-loop strategies and overall task performance drops sharply.

Predictive displays and simulation is one approach to predict the state of the remote vehicle environment and present that prediction to the operator to provide a sense of instantaneous feedback [107]. If there is a half-second delay, the system can extrapolate the vehicle's motion or use a local simulation so that the operator sees a predicted vehicle position that is more up-to-date [82]. Early NASA telerobotics experiments used this method to let operators drive a simulated model of the rover, which would then execute the commands effectively and the operator experiences zero delay in their own loop, though the real robot lags behind.

Time-delayed control remains a rich research domain because internet-based teleoperation will always have non-zero latency. A recent survey [108] specifically on network latency in connected autonomous vehicles teleoperation compiles state-of-the-art compensation approaches and emphasizes that both control-side solutions and network-side solutions are needed for seamless integration of teleoperation into vehicle systems. One trend is the use of machine learning to handle delay [109], [110]. For example, training a neural network to predict the operator's next action or the vehicle's next state and thereby buffer the delay [111], [112]. These data-driven predictive methods are gaining popularity as they can be more adaptable to varying conditions.

2) *Predictive Control*: Model Predictive Control (MPC) is an optimization-based control strategy that computes control actions by predicting future vehicle states over a moving horizon [113]. At each time step, an MPC controller solves a constrained optimal control problem to minimize tracking error while satisfying system constraints such as actuator limits. This approach means only the first input of the optimal sequence is applied before the optimization is repeated at the next step. MPC's ability to explicitly handle multivariable dynamics and constraints without ad-hoc tuning of complex control laws makes it a compelling choice for advanced vehicle control. In the context of vehicles, MPC commonly uses a simplified predictive model (e.g. a kinematic bicycle model) for real-time feasibility, and it naturally accommodates the vehicle's physical limits and performance objectives within its cost function and constraints.

3) *Assistive Control*: Assistive control in semi-autonomous teleoperation blends human inputs with AI-driven guidance to enhance safety and reduce operator workload. Assistive control in semi-autonomous teleoperation augments human commands with AI-driven assistance to enhance safety and reduce operator workload. Steering-action-aware controllers apply corrective torques when sensors detect imminent obstacles, guiding the operator's inputs away from hazards without fully overriding manual control [114]. POMDP-based shared-control frameworks fuse a belief over the environment and operator intent to optimally blend human and autonomous commands under uncertainty [115]. Model Predictive Control combined with Control Barrier Functions enforces safety constraints on user inputs intervening only when state trajectories threaten to violate safe margins thereby avoiding oscillations or instability in delayed networks [116]. Deep reinforcement-learning-based assistive policies further adapt the level of autonomy in real time based on network conditions and driving context, ensuring smooth assistance under variable delays [110]. Field trials of active safety systems for teleoperated vehicles demonstrate that these assistive approaches significantly reduce collision rates and operator cognitive load [117]. Altogether, these assistive control techniques provide a scalable and reliable teleoperation architecture that balances human judgment with autonomous safety to enable practical deployment of semi-autonomous vehicles.

IV. AUTONOMY

Teleoperation of semi-autonomous vehicles represents an evolving frontier in the field of autonomous driving where the human operator may intervene when necessary while the vehicle handles some driving tasks autonomously [118]. The taxonomy of autonomy in teleoperation is generally categorized based on the levels of human involvement in vehicle control and the vehicle's capability to operate independently. A widely used standard for classifying autonomy is provided by the SAE International J3016 [77] shown in Table II, which defines levels of automation from Level 0 (no automation) to Level 5 (full automation). These levels are fundamental for understanding the interaction between human operators and autonomous systems in teleoperation.

A. Decision Making Methods

AI algorithms are crucial for improving the functionality, safety and responsiveness of teleoperated semi-autonomous vehicles. A variety of AI algorithms have been applied to improve decision-making in teleoperation systems, ranging from machine learning methods to probabilistic models [119], [120].

Reinforcement Learning (RL) has been used [121] to develop shared control policies where an autonomous agent learns to assist or complement the human operator's commands. For example, a teleoperation framework using a truncated quantile critics deep RL agent was shown [122] to significantly improve task completion time, success rate and reduce operator workload in a remote manipulation task. The RL agent learns when to intervene or guide the human, resulting

in a human-AI team that outperforms manual teleoperation alone.

Deep neural networks contribute to teleoperation in multiple ways. Vision-based deep learning models are used for scene perception and understanding, helping the remote operator by automatically detecting obstacles or hazards and suggesting actions [122]. Recurrent neural networks (RNN) have been employed to predict operator commands or vehicle states in advance, which is useful for smoothing control and compensating latency [129]. For instance, long short-term memory (LSTM) networks can learn temporal patterns in driver inputs and effectively mitigate control delays by predicting the teleoperator's next action [130]. Such learned predictive models enable the vehicle to anticipate human decisions, resulting in more responsive and stable teleoperation.

In addition to learning-based approaches, probabilistic decision models help manage uncertainty in teleoperation. Bayesian reasoning [131] and state estimation [132] are often used to fuse sensor information and human inputs. For example, teleoperated unmanned ground vehicles can build a local occupancy grid map using probabilistic state estimation, then compute navigable paths within that map. Cognitive models from human factors research have also been leveraged where one study used a two-point visual steering model and an ACT-R cognitive architecture to predict a teleoperator's steering behavior [133]. By modeling the human's likely actions, the system can assist or correct the course proactively. These probabilistic and cognitive approaches enhance decision-making by accounting for uncertainties in both environment perception and human behavior.

B. Latency Mitigation and Safety Enhancement Methods

AI models can predict future states or operator commands to bridge communication delays [134], [135]. Time-series prediction methods like LSTM [109] or GRU [136] neural networks have proven effective for delay compensation. For example, a teleoperated robot can run an LSTM-based predictor that estimates the next control input from the human, allowing the vehicle to start reacting before the actual command arrives. Advanced implementations even combine learning with control theory where one study integrated an RNN predictor with a deep-Q reinforcement learning agent to anticipate an operator's actions in real-time [137]. These predictive techniques smooth out jerky motions and maintain stability, essentially masking moderate network delays so that teleoperation feels more instantaneous. In commercial remote driving platforms, intelligent algorithms predict wireless link quality and adjust video encoding or control update rates accordingly. An example is Imperium Drive's system [138], which uses AI to forecast network latency/packet loss and proactively downgrade to safe control modes if the connection degrades. By anticipating drops in bandwidth or high latency, the system can reduce vehicle speed or enter a precautionary state, thus maintaining safety. Such teleoperation powered by AI ensures the vehicle remains under control even in

TABLE II
SAE LEVELS OF DRIVING AUTOMATION INCLUDING TELEOPERATION

Level	Description	Human Role	Example
0	No Automation	The human operator is fully responsible for all aspects of vehicle operation.	Manual driving with no assistance systems
1	Driver Assistance	The system may assist with specific tasks like steering or acceleration, but the human operator is still responsible for overall control.	Adaptive cruise control or lane-keeping assistance [123]
2	Partial Automation	The vehicle can control both steering and acceleration, but the human operator must remain engaged and ready to take control.	Tesla Autopilot under supervision [124]
3	Conditional Automation	The vehicle can perform most driving tasks, but the human operator must be ready to intervene if necessary.	Audi's Traffic Jam Pilot [125]
3.5	Semi-Automation	The Vehicle performs driving tasks Autonomously, but the human operator intervenes remotely when necessary.	[126]
4	High Automation	The vehicle can operate autonomously within a defined operational domain without human intervention.	Autonomous shuttles in urban environments [127]
5	Full Automation	The vehicle can operate autonomously in all environments without human intervention.	Fully autonomous vehicles [128]

fluctuating 4G/5G network conditions.

To guarantee safety, many teleoperated vehicles incorporate on-board autonomous interventions that are triggered by AI [114], [139]. A semi-autonomous controller can continuously monitor the vehicle's sensors and override or correct the human's commands if an imminent collision or hazard is detected. Research has demonstrated this approach in remote driving where if the teleoperator's steering input would lead the car toward an obstacle, the vehicle's AI will adjust the steering to avoid it [140]. Importantly, the autonomous safety layer only intervenes when necessary like when braking for an unseen obstacle or preventing a dangerous maneuver so as not to wrest control from the human except in critical moments [141]. This guarantees a fail-safe operation despite latency or human error. The AI acts as a guardian that can react faster than the remote human in emergency scenarios.

Beyond prediction and intervention, AI algorithms grounded in control theory like model-predictive control combined with learned models help maintain stable teleoperation [142]. For example, tele-driving researchers have combined MPC with Kalman filters to forecast vehicle state and compensate for delays, thereby avoiding oscillations or instability [143]. Others employ passivity-based or adaptive controllers augmented with neural networks to guarantee stability even with variable time lags [144]. While not all methods use learning, many modern approaches hybridize classical control with AI components like neural network function approximators to handle uncertainties. The result is teleoperation systems that remain robust under high latency and can recover gracefully from brief connection losses.

AI-driven latency mitigation allows teleoperation to be viable over long distances and wireless networks. By predicting operator intent, adapting to network changes and adding autonomous safety nets, these approaches address the inherent

delays and unreliability of remote communication [145]. In practice, a combination of such methods is often used to ensure that teleoperated semi-autonomous vehicles respond promptly to commands and avoid accidents, even when faced with the inevitable latency of real-world networks.

C. Autonomy in Multi-Robot Systems

Multi-robot teleoperation involves a human operator remotely controlling or supervising a team of robots or vehicles [146], [147]. Such systems have gained prominence alongside the rise of semi-autonomous vehicles, which can perform tasks independently but still benefit from human intervention. A key motivation is to leverage human expertise to handle complex or unexpected situations, thereby improving overall mission performance while keeping humans out of harm's way. For example [148], autonomous mobile robot teams in manufacturing can function independently but still call upon a remote human for assistance when facing challenges, combining the efficiency of autonomy with the adaptability of human guidance. This synergy is especially important for semi-autonomous vehicles that operate in dynamic or safety-critical environments where full autonomy may not be reliable.

Coordinating multiple robots under teleoperation requires sophisticated control strategies to prevent the operator from being overwhelmed by micromanagement [149]. For example, leader-follower formations or consensus-based movements such that the operator can issue a single command and have all robots respond in a organized way [150]. In practice, this might mean one robot is directly teleoperated (the lead) while the others automatically follow its trajectory or maintain a formation, drastically reducing the operator's workload. Researchers have shown [118] that if the control system can ensure follower robots track the actions of a lead vehicle, a single operator can effectively drive an entire

convoy with minimal additional effort. Beyond formations, task allocation algorithms play a vital part in coordination, when multiple robots have to achieve a set of objectives, the system can autonomously distribute tasks among the robots and cue the human operator only when intervention is needed for a specific task. This concept is embodied in sliding or adjustable autonomy, where robots handle routine subtasks and the human concentrates on the critical decisions. Recent work has framed multi-robot teleoperation coordination [151] as an optimization and scheduling problem. For instance, one study [148] proved that deciding which robot and task an operator should teleoperate at any given time to minimize overall mission duration is NP-hard. To tackle this, the authors formulated a Mixed-Integer Linear Program that generates optimal schedules for operator attention and also developed an anytime algorithm for larger problem instances. The resulting decision support system produces a schedule telling the operator which robot to control and when, ensuring that each robot either advances autonomously or waits for assistance in a logical sequence. Such coordination algorithms prevent bottlenecks and maximize team efficiency under single-operator constraints. In summary, multi-robot coordination control blends classical multi-agent control theory with unique teleoperation considerations. By offloading low-level coordination to algorithms having robots sync their movements for the operator's attention. These methods enable human operators to focus on high-level guidance. This division of labor between autonomous coordination and human decision-making lies the foundation for scalable multi-robot teleoperation.

V. HUMAN FACTORS IN SEMI-AUTONOMOUS TELEOPERATION SYSTEMS

Human factors play a critical role in the effectiveness and safety of semi-autonomous teleoperation systems. In many applications, remote human operators are required to supervise or control vehicles, especially when full autonomy is limited by technical or environmental constraints. Factors such as trust, cognitive workload, situational awareness (SA) and interface usability directly influence operator performance and decision-making. These challenges become more significant when supervising multiple vehicles or switching between control and monitoring modes [14], [84], [152]. This section discusses how human factors intersect with communication, control, and autonomy in teleoperated systems, highlighting design implications and evaluation approaches. Specifically, this section examines the role of human factors across three interrelated domains: (1) single-vehicle teleoperation under varying network performance conditions, (2) the cognitive and operational demands of supervising multiple autonomous vehicles, and (3) the complexities of mixed-initiative control where authority is dynamically shared

A. Human Factors and Network Performance

Remotely operating semi-autonomous vehicles, whether in everyday scenarios or extreme environments, demands a robust communication infrastructure and effective human-machine

interaction. This is especially critical when vehicles are deployed in hazardous or high-risk contexts such as nuclear maintenance, planetary exploration, or industrial disaster response [153]. In these scenarios, the success and safety of remote driving (teleoperation) hinge on the interplay between communication parameters, including latency, bandwidth, and video quality, and key human factors such as cognitive workload, situational awareness, trust, and overall user experience. Understanding how communication impairments impact operator performance is essential for designing resilient, human-centered remote driving systems that can adapt across diverse conditions. Table III summarizes representative studies that investigate how network performance affects human factors during the teleoperation of semi-autonomous vehicles, particularly in complex or safety-critical scenarios. These studies collectively offer insights into performance outcomes, operator adaptation strategies, and the role of supportive interface technologies.

Across studies involving both simulation and real-world operators [9], [10], [54], [107], [154], [156], [158], elevated latency is reliably associated with degraded task performance, increased subjective workload (as measured, for example, by the NASA-TLX instrument [10]), and diminished situational awareness. For example, Neumeier et al. [9] found that latencies of about 300 ms and above led to deteriorated driving performance and decreased operator confidence in teleoperated driving. Similarly, studies involving unmanned ground vehicles [10], [158] and simulated robotics platforms [54], [154] further show that delay-induced increases in workload are often more apparent in subjective ratings than in objective metrics, highlighting challenges unique to human-centered teleoperation.

Degraded video quality and limited field of view [11], [156] exacerbate these issues, especially under high-contrast or visually complex conditions. Rafiei et al. [156] found that both reduced resolution and increased latency led to higher error rates and lower operator comfort during remote control of industrial equipment. Mitigation strategies, including delay compensation aids [10], [158], predictive displays [107], and enhanced interface designs [54], [154], are shown to partially improve performance and operator comfort, though none fully eliminate the adverse effects of network constraints under significant delay. Avatar-aided and predictive approaches, in particular, have been found to reduce subjective workload and improve perceived safety in high-latency remote scenarios [107], [154].

Qualitative and interview-based research [155], [157] complements these experimental findings by highlighting that operator trust in the remote system depends not only on real-time network quality but also on operator training, system transparency, and adaptive strategies. For instance, Louca et al. [155] report that in high-risk domains such as nuclear maintenance and bomb disposal, operator trustworthiness is rooted in thorough system understanding and experience with multimodal feedback, especially when high or variable latencies require different operator command strategies.

Taken together, these studies demonstrate an explicit, empirically supported link between communication metric degra-

TABLE III
COMPARISON OF KEY STUDIES ON COMMUNICATION METRICS, HUMAN FACTORS, AND TELEOPERATION IN HAZARDOUS ENVIRONMENTS.

Communication Metric(s) Studied	Human Factor(s) Measured	Context / Vehicle Type	Key Findings	Reference
Network latency (constant, variable; 300ms+)	Workload, confidence, task performance	Automated vehicle teleop (simulator)	Increasing latency degrades performance, increases workload, reduces operator confidence. Variable latency did not always worsen performance compared to constant.	Neumeier et al. (2019) [9]
Video latency (3s)	Workload, usability, robot safety	Planetary exploration robot (virtual)	Avatar-aided and predictive interfaces significantly reduce workload and improve usability and robot safety versus baseline under high latency.	Luz et al. (2023) [154]
Time delay (0.8s), delay compensation aid	Workload (NASA-TLX), performance	Unmanned Ground Vehicle (UGV), dual-task sim	Delay increases workload and degrades performance. Model-free predictor compensation aid mitigates these effects. Subjective workload is more sensitive than objective and physiological measures.	Lu et al. (2019) [10]
Video delay (900/1200ms), video transformation aids	Task completion time, Situational Awareness (feedback), controllability	UGV, on/off-road sim	Latency alone greatly increases task completion times and decreases controllability. Video transformation aids improve visual Situation Awareness and operator comfort.	Moniruzzaman et al. (2022) [54]
Camera/LiDAR delay (0.5s), predictive display	Teleoperation performance	Automated Guided Vehicle (AGV)	0.5 s camera delay makes operation extremely difficult. Predictive display using model-based estimation restores usability.	Sharma & Rajamani (2024) [107]
Signal latency (up to 15s, domain dependent)	Trust, Situational Awareness, cognitive strategies	Expert interviews: nuclear, surgical, underwater, bomb disposal	Trust depends on system understanding, transparency, multi-view feedback, and operator experience. High/variable latency requires adaptive, slower, or higher-level commands.	Louca et al. (2023) [155]
Video quality, latency	Field of view, User experience, comfort, performance error	Industrial remote heavy vehicle (laboratory setting)	Increased latency and lower video quality increase error rates and reduce operator comfort.	Rafiei et al. (2023) [156]
Latency (> 150ms), bandwidth, reliability	Workload, SA, adaptation/trust (survey, interview)	Commercial remote driving operators	Latency and reliability remain challenges. Operators adapt but report lower trust/confidence compared to in-vehicle driving; training and adaptation are ongoing issues.	Hardin et al. (2024) [157]
Time delay, delay compensation aid	Workload, performance (dual-task)	Simulated High Mobility Multipurpose Wheeled Vehicle (HMMWV), UGV teleop	Delay compensation aid reduces workload and improves primary task performance under delayed conditions.	Lu et al. (2018) [158]

ditions and declines in the human factors crucial for safe and effective teleoperation of semi-autonomous vehicles in hazardous environments. The literature indicates that while technical and interface-based mitigations are important, operator training and transparent system design remain key to sustaining trust and situational awareness under non-ideal communication conditions.

Synthesizing these findings, a recent systematic review by Kamtam et al. [153] further highlights that the impact of network latency extends beyond performance degradation to impose significant cognitive, perceptual, and behavioral demands on remote drivers. The study emphasizes that many existing teleoperation systems fall short of supporting ideal user experiences, particularly under variable or high-latency conditions. It calls for a more integrated human-centered design approach that aligns system performance metrics, such as uplink/downlink latency and data throughput with the real-world cognitive and psychological requirements of human

operators. This reinforces the view that optimizing human factors in teleoperation is not simply a usability issue, but a safety-critical imperative.

This perspective is further supported by the work of [48], [159], who found that beyond latency and interface mitigation, deeper challenges remain. Situational awareness limitations are particularly prominent, as remote operators often lack sufficient spatial context, depth perception, and multi-sensory cues. Effective teleoperation requires enhanced perception support through tools such as map integration, viewpoint adjustment, and haptic or audio feedback. However, poorly designed systems risk introducing cognitive overload, distraction, or loss of control. Additional constraints, such as lack of physical sensing, unreliable control under uncertainty, bandwidth bottlenecks, and insufficient local autonomy, further compound the issue [48], [159]. Together, these factors emphasize the urgency of developing holistic teleoperation systems that are not only network-efficient but also carefully adapted to the

TABLE IV
NETWORK AND RELATED INTERFACE ISSUES, HUMAN FACTOR IMPACTS, AND MITIGATION STRATEGIES IN SEMI-AUTONOMOUS VEHICLE TELEOPERATION

Network and related interface Issue	Effect on Human Factors	Suggested Mitigation Strategies
High Latency (e.g., 300ms)	Increased cognitive workload, delayed task performance, reduced operator confidence	Delay compensation algorithms, predictive displays, proactive command buffering
Low Bandwidth	Reduced system responsiveness, increased frustration, decreased trust	Adaptive streaming, efficient encoding, prioritization of control signals
Poor Video Quality	Reduced situational awareness, visual perception errors, slower decision-making	High-resolution video, panoramic or multi-camera views, immersive HMDs
Unstable Connectivity	Higher stress, control difficulties, degraded overall experience	Connection redundancy, stable communication protocols, real-time QoS monitoring
Camera/LiDAR Delay	Decreased responsiveness, higher mental demand, poor controllability	Predictive visual displays, model-based estimation, latency buffers
Limited Field of View	Reduced spatial awareness, navigation errors	Wide-angle cameras, 360° visualization, improved field of perception
Packet Loss / Jitter	Disrupted feedback loop, command misinterpretation, trust degradation	Error correction, jitter buffering, redundant data pathways
Audio/Visual Delay or Loss	Mistimed decisions, environmental confusion, increased workload	Synchronized AV streams, audio cues, alert systems with low latency
Lack of Multisensory Feedback	Low immersion, difficulty interpreting remote cues, lower user engagement	Haptic feedback (vibrations, force cues), audio feedback (spatial alerts)

operator's cognitive, sensory, and decision-making needs.

Based on the reviewed literature and experimental findings discussed in this section, it is possible to distill a set of recurring patterns linking network communication issues to critical human factors in the teleoperation of semiautonomous vehicles. Table IV provides a summary of these relationships. It identifies the most commonly reported network-related challenges, outlines their observed or inferred impacts on human cognitive, perceptual, and behavioral dimensions, and presents mitigation strategies that have been proposed or evaluated in prior research. This synthesis aims to offer a practical reference framework for understanding and addressing human-centered risks in remote driving scenarios where communication impairments are likely to occur.

B. Human Factors in Multi-Vehicle Teleoperation

While the challenges of single-vehicle teleoperation are well documented, the complexity increases significantly in multi-vehicle scenarios, where cognitive demands, coordination requirements, and trust dynamics are further amplified [152], [160], [161]. As autonomous robotic systems become increasingly capable, operational paradigms have shifted from direct teleoperation of individual robots to supervisory control of multiple autonomous agents by a single human operator. This transition aims to improve system scalability and resource efficiency in domains such as surveillance, infrastructure inspection, and emergency response [13], [152], [162]. However, increasing the operator-to-vehicle (O:V) ratio introduces significant challenges for human cognitive workload, situational awareness, and effective exception handling. These challenges require careful design and evaluation of human-robot interfaces and autonomy management strategies to ensure that operators can maintain adequate oversight, intervene when

necessary, and avoid performance breakdowns under high workload or in the face of automation surprises.

Recent findings from a controlled study on remote supervision of autonomous vehicles further underscore these challenges, particularly when scaling operator responsibilities across multiple vehicles [12]. The study revealed that operators maintained a consistent ability to detect automation failures when managing one or two vehicles, though this came with increased cognitive workload. However, performance began to degrade noticeably when four vehicles were introduced, suggesting a potential upper limit to effective human supervision in such contexts. Additionally, a decline in takeover accuracy over time indicated the impact of fatigue or vigilance decrement, even when self-reported workload levels remained stable. These insights highlight critical design considerations for supervisory control systems in autonomous driving applications, including operator-to-vehicle ratios and shift duration limits to ensure timely and accurate human intervention during unexpected events.

Moreover, experimental and human-in-the-loop studies have systematically examined the cognitive and operational limits inherent in multi-robot supervision. Researchers have employed controlled laboratory and simulation-based experiments to compare traditional graphical interfaces, immersive VR systems, and predictive display elements, often manipulating the number of robots supervised and measuring outcomes using subjective workload (e.g., NASA-TLX) and situational awareness probes (e.g., SAGAT) [13], [152], [163], [164]. Notably, [152], [163] demonstrated that immersive VR and predictive interfaces can significantly reduce key workload components and improve operator SA during supervisory tasks with up to three robots. Similarly, [164] found that multi-modal communication cues enhance situational awareness and responsiveness, particularly as the O:V ratio increases.

Further research has addressed the impacts of interruptions and task switching on multi-robot supervision. Authors in [13] conducted a human-in-the-loop study varying the O:V ratio (1:4 vs. 1:9) and the type of interruption (fault correction vs. messaging), revealing that performance declines with increasing numbers of robots and that interruptions, especially extrinsic ones, elevating perceived workload. Adaptive and attention-sensitive autonomy systems have been explored as a potential solution to workload management. For example, authors in [162] introduced a gaze-driven adaptive autonomy model which dynamically balances operator attention across four robots, resulting in improved operational performance.

Several studies have investigated the trade-offs associated with increasing autonomy levels, management-by-exception, and adaptive workload allocation. Studies [14], [165], [166] highlighted that while advanced decision support and automation can reduce manual demand, they may also produce new risks, such as operator fixation and situational awareness degradation, if not properly designed and calibrated. Additional work has examined physiological and performance-based workload assessment [167], [168], the benefits of group autonomy over individual robot control [168], and the role of explicit alerts and attention management cues in supporting scalable supervision [164], [169].

Recent frameworks, particularly relevant to Level 3.5 systems involving shared control responsibilities, have proposed a more comprehensive classification of remote operator roles in highly automated vehicles, distinguishing between remote monitoring (RMO), remote assistance (RA), remote management (RMA), and remote driving (RD), each aligned to different interaction levels with the ADS [170]. These distinctions acknowledge that remote operators must flexibly transition between roles depending on the scenario, and that situational awareness, anticipation, and role-based decision making are central to effective remote intervention. Importantly, in SAE Level 3 vehicles, where automation handles driving under defined conditions but requires human fallback intervention, remote operation plays a crucial supervisory role. In this context, a *remote fallback-ready user* must be capable of assuming partial or full control upon system request, making real-time responsiveness and cognitive readiness essential to safety and mission continuity. As highlighted by [171], reliable remote operation in Level 3 scenarios hinges on network performance, particularly ultra-low latency to enable timely handovers, and the design of human-machine interfaces that maintain operator readiness without inducing fatigue.

Collectively, these studies converge on the conclusion that effective multi-robot supervision requires interfaces and autonomy strategies that proactively manage cognitive workload and maintain situational awareness, especially as O:V ratios rise beyond empirically supported limits (often 1:4–1:8). The literature underscores the importance of adaptive, immersive, and multi-modal supports, alongside careful consideration of operator trust and involvement in intervention and exception handling, to enable robust and scalable human supervision of autonomous robotic systems [13], [14], [152], [162]–[165], [169].

From a human-centred perspective, it is essential that remote

operators, particularly in Level 3.5 scenarios, maintain a clear understanding of their current role within the shared control context. Due to the dynamic nature of supervision, operators may transition between monitoring, assisting, or directly intervening, and ambiguity in role awareness could impair reaction time or decision quality [172]. Therefore, interfaces should be designed to explicitly indicate the operator's current level of involvement and responsibilities. Visual cues integrated into head-mounted displays or immersive VR [163] environments can support this by reinforcing role clarity and situational awareness.

Furthermore, intelligent interface systems that incorporate artificial intelligence may enhance this interaction by predicting when role transitions are likely and proactively alerting the operator [173]. Such systems can also assess performance, workload, and situational context to recommend appropriate levels of intervention or rest. These mechanisms not only facilitate smoother coordination between human and machine, but also reduce cognitive ambiguity during time-sensitive events.

In addition, dedicated training programs for remote fallback-ready drivers are critical. Training should address not only task execution under normal conditions, but also how to interpret interface signals, understand automation limitations, and manage role-switching effectively [171]. This will ensure that remote operators are prepared to handle high-demand supervision tasks with confidence, clarity, and safety. Given the pivotal role that interface design plays in maintaining operator awareness and managing cognitive workload in remote supervision scenarios, it is essential to evaluate the relative effectiveness of different display modalities. In particular, the following section compares immersive VR head-mounted displays and traditional screen-based interfaces in the context of semi-autonomous vehicle teleoperation.

C. Comparing VR Head-Mounted Displays against Traditional Monitors

The supervision and teleoperation of semi-autonomous vehicles demand interfaces that optimally support human situational awareness, cognitive workload management, vigilance, and overall user experience. Recent advancements have catalyzed interest in immersive head-mounted displays (VR HMDs) as alternatives to traditional TV/monitor-based setups. However, the comparative efficacy of these modalities, especially in real-world or ecologically valid teleoperation scenarios, remains an active research question.

A series of empirical studies directly compares VR HMDs and multi-monitor or screen-based interfaces for remote vehicle supervision and teleoperation. Notably, [cite kalamkar2023remote](#) presents controlled user studies finding that while VR HMDs increase subjective presence, they can lead to higher cognitive workload, increased simulator sickness, and no clear performance improvement compared to physical multi-monitor configurations. Similarly, [16] contrasts HMDs and conventional screens during urban vehicle teleoperation, reporting heightened immersion with VR but no reduction in collisions or demonstrable gains in control.

TABLE V
SUMMARY COMPARISON OF TOP EMPIRICAL VR HMD VS. SCREEN-BASED DISPLAY STUDIES FOR VEHICLE TELEOPERATION.

Application	Main Measures	Key VR HMD Findings	Screen/Monitor Findings	Reference
AV monitoring / teleoperation	Alert misses, RT, TLX, SUS, IPQ, SSQ	Higher presence and task load, more alert misses, increased sickness symptoms	Faster alert reactions, lower workload	[15]
Real-vehicle teleoperation	Driving performance, SA, immersion	Greater immersion, no performance benefit, some collisions	Baseline performance, no immersion gain	[16]
Remote assistance, 360° video	Decision speed, accuracy	Faster and more accurate decisions; some comfort concerns during prolonged use	Slower decisions, lower accuracy	[17]
Weeks-long monitoring	SA, workload, presence, usability	Preferred subjectively; no significant advantage in SA or workload	Similar objective metrics; less user preference	[11]
UGV teleoperation	Navigation, workload, usability	Better spatial accuracy, reduced workload, higher usability ratings	Lower performance in navigation tasks	[174]
UGV teleoperation	SA ? (SAGAT), navigation	Enhanced SA and navigation efficiency	Reduced SA and operational efficiency	[175]

Complementary work has explored related decision-making and situation awareness tasks. [17] demonstrate that 360° video presented in a VR HMD improves decision speed and accuracy compared to a screen-based interface in remote assistance tasks, albeit cautioning regarding potential user discomfort during prolonged use. Longitudinal evaluation by [11] reveals that, over several weeks, participants prefer HMDs, though objective performance metrics such as situation awareness and workload remain comparable across display types. Research in closely allied domains, such as UGV teleoperation, indicates that immersive stereoscopic VR presentations can significantly enhance navigation accuracy, workload, and situation awareness compared to 2D monitors, especially for spatially demanding tasks [174], [175].

Together, these studies illuminate a nuanced landscape: VR HMDs may offer advantages for spatial decision-making and immersive engagement, but their benefits do not consistently extend to prolonged monitoring or vigilance-heavy tasks in semi-autonomous vehicle teleoperation. Table V summarizes key findings from the most directly relevant empirical comparisons.

Despite significant progress, several key gaps and challenges remain explicit in the current literature, indicating clear avenues for future investigation:

- **Trust Calibration:** Most controlled comparisons to date have focused on performance, workload, and presence, with little direct measurement of operator trust or trust calibration in automation. Future studies should include validated trust instruments and explore how display modality influences reliance and intervention timing [15]–[17].
- **Long-Term Fatigue and Simulator Sickness:** While short-term usability and workload have been assessed, there is a need for research on the long-term effects of VR HMD use, including fatigue, discomfort, and sustained vigilance, especially over extended monitoring or supervisory teleoperation periods [11].
- **Hybrid and Adaptive Interfaces:** Early work on AR/VR overlays and context-adaptive displays suggests potential

for combining advantages of immersive and traditional modalities, but empirical evaluation of such hybrid systems is still nascent [176].

- **Domain Transferability and Task Specificity:** Further research should clarify the generalizability of findings from UGV/drone teleoperation to road vehicle contexts and dissect which sub-tasks (e.g., obstacle avoidance, alert monitoring) benefit most from VR or screen-based interfaces [15], [17], [174], [175].
- **Methodological Improvements:** Future work would benefit from larger participant samples, systematic reporting of both objective and subjective measures across varying levels of operator experience, and head-to-head comparisons that include both performance and user experience endpoints.

D. Human-Autonomy Interaction in Mixed-Initiative Teleoperation

Beyond general human-Autonomy coordination, a growing body of research investigates more nuanced models of shared authority, known as mixed-initiative control [177]. Mixed-initiative teleoperation, in which human operators and robots dynamically share control of semi-autonomous ground vehicles, presents significant Human-Computer Interaction (HCI) challenges, particularly concerning authority handover, real-time coordination, and the maintenance of situational awareness. Recent empirical research substantiates the value of adaptive and dynamic handover mechanisms, transparent interfaces, and user-centered procedures in these systems [177].

Robust, real-time mixed-initiative frameworks such as the Expert-guided Mixed-Initiative Control Switcher (EMICS) have enabled both humans and robots to dynamically shift authority based on performance metrics and contextual rules [177]. Usability studies using EMICS in real and simulated disaster-response tasks quantified reductions in operator workload (NASA-TLX), increases in trust, and improved navigation performance compared to single-mode or operator-driven switching, highlighting interface designs that support live map/video overlays and manual control of levels of

autonomy [177]. Building on this, negotiation-based models (e.g., Negotiation-Enabled Mixed-Initiative Control Switcher (NEMICS) algorithm) introduce explicit, bilateral handover dialogue and interface notifications, resulting in fewer control conflicts, improved perceived safety, and lower user-reported intrusiveness [178].

Adaptive schemes increasingly integrate real-time assessments of operator state, such as cognitive availability via computer vision [179], behavioral entropy [180], or physiological and behavioral signals [181], to optimize moment-to-moment authority allocation and trigger interface feedback or alerts. Advanced trust-modeling strategies, using Kalman filtering or Bayesian relational-event models, enable systems to modulate autonomy levels responsively, which empirical studies show can further reduce operator workload and enhance system usability [182], [183]. Recent work also explores interface innovations such as workload-adaptive haptic guidance [184], dynamic warning panels, and hierarchical arbitration of handover triggers [185]. Multimodal feedback and explicit authority widgets are found to support smoother and more efficient human-Autonomy collaboration [177], [178], [184], [185]. There is a lack of standardized handover protocols or longitudinal, real-world evaluations, with most research relying on short-duration laboratory or simulated scenarios [177], [186], [187].

In summary, the literature demonstrates steady progress in the development and empirical validation of adaptive, mixed-initiative control architectures and user-adaptive teleoperation interfaces. However, richer transparency mechanisms, robust assessment of situational awareness during handover, and standardization of handover protocols remain key frontiers for future research.

VI. TRAINING HUMAN OPERATORS FOR SEMI-AUTONOMOUS VEHICLE SYSTEMS

Semi-autonomous vehicle systems are becoming increasingly integrated into modern transportation. Despite this automation, human operators still play a critical role, particularly in supervisory control, during system limitations, and when transitioning between automated and manual driving modes. Training programs for such systems must be carefully designed to develop not only technical proficiency but also cognitive and behavioral preparedness for effective human-automation interaction. Human factors such as situational awareness, trust calibration, mental workload, and cognitive flexibility have been shown to significantly influence operator performance and safety. According to [172], failures in automation contexts are often linked to reduced situational awareness and misaligned levels of trust, highlighting the need for training approaches that explicitly address these variables. Consequently, effective training should be based on validated models of human factors in automation to ensure that operators are equipped to manage both routine interactions and unexpected edge-case scenarios.

This section discusses best practices and emerging approaches in training protocols for remote operators and safety drivers of semi-autonomous vehicles, including immersive

simulation, VR-based training modules, and scenario-driven exercises that enhance operator readiness under variable network and system conditions. The section also explores how performance metrics such as response time, decision accuracy, and trust calibration evolve over repeated training exposures, contributing to operator adaptation and resilience in real-time operation contexts.

Recent research has focused on the effectiveness of immersive training modalities, including VR, AR, and their application in both simulation and live operational settings. Authors in [188] and [18] present controlled comparisons between light, portable VR training, traditional user manuals, and high-fidelity simulators, evaluating the transfer of training to unplanned take-over requests (TORs) in semi-autonomous vehicles. These studies report that VR-based approaches achieve takeover reaction times and control quality on par with fixed-base simulators and surpass traditional methods in scalability and user preference.

Augmented reality interfaces have also demonstrated significant potential. [189] and [190] find that AR overlays can improve lateral vehicle control, reduce subjective workload, and increase situational awareness when compared to conventional displays. Digital twin visualizations and stereoscopic 3D interfaces have further shown benefits in early takeover preparation by enhancing driver awareness of upcoming scenarios before handover events [191].

The integration of adaptive and multimodal feedback mechanisms, such as visual, auditory, and vibrotactile cues, has been evaluated for their impact on takeover performance, workload, and trust. Studies by [20] and [192] highlight that visual signals, as well as directional vibrotactile cues, can significantly reduce takeover times and increase both situational awareness and trust.

Beyond interface and simulator studies, knowledge-based training frameworks have been proposed to address cognitive readiness. Authors in [19] and [193] demonstrated that explanation-based knowledge of system limitations and handover concepts increases both the rate of successful interventions and reduces response times, particularly for first-time and older operators.

Despite this progress, there remains a scarcity of research integrating multi-phase curricula that progress from foundational VR drills to live, mixed-reality vehicle training for edge-case practice. Furthermore, standardized quantitative assessment of situational awareness and cognitive readiness, particularly under live or high-fidelity conditions, is limited. The current literature thus provides foundational evidence for the efficacy of immersive, human-factor-targeted operator training but underscores the need for further research combining these modalities in operational environments.

VII. CONCLUSION

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This should be a simple paragraph before the References to thank those individuals and institutions who have supported your work on this article.

APPENDIX

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